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Course:
Intro to Computing
Cs-101

BS Information Technology
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Natural Language Processing

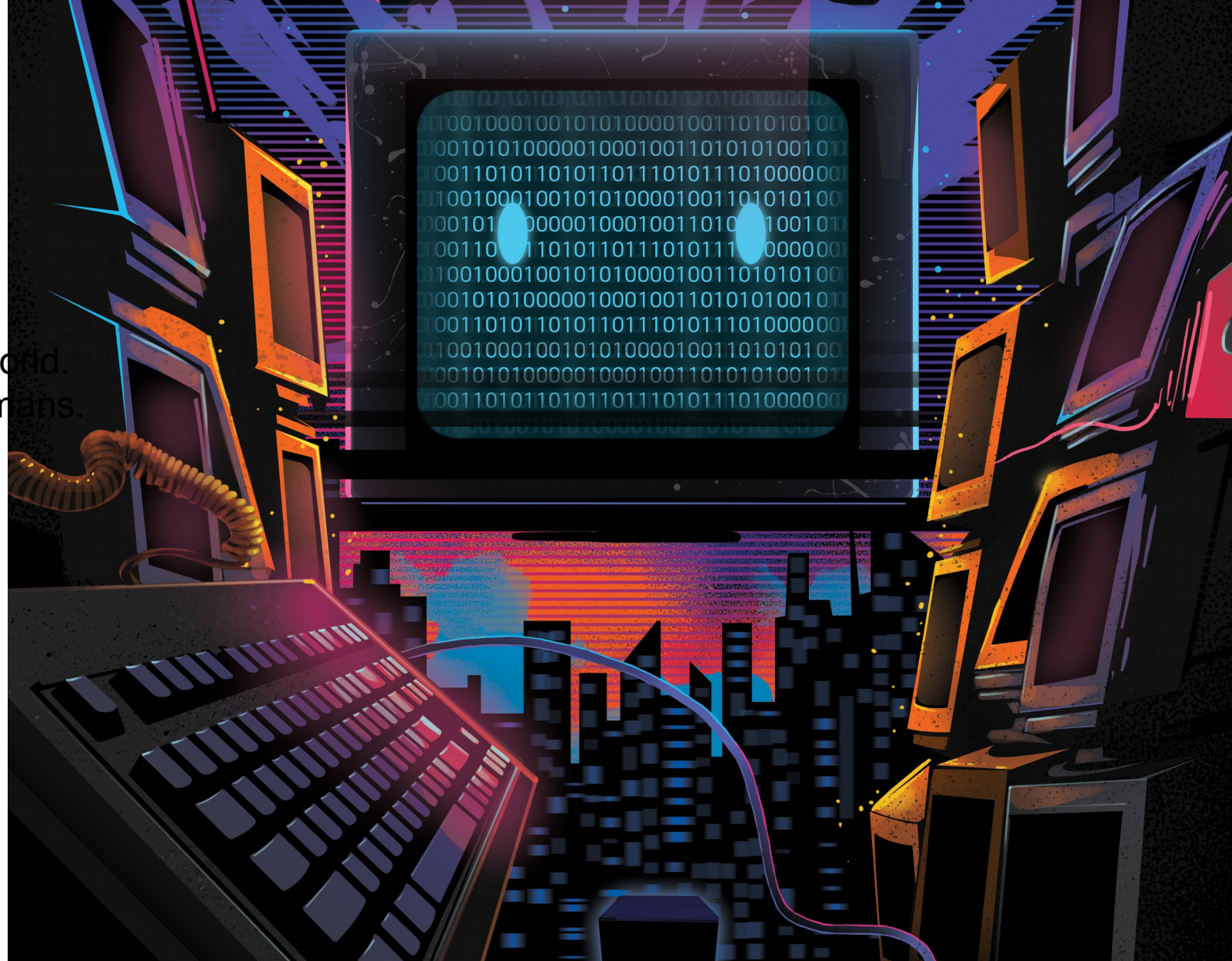
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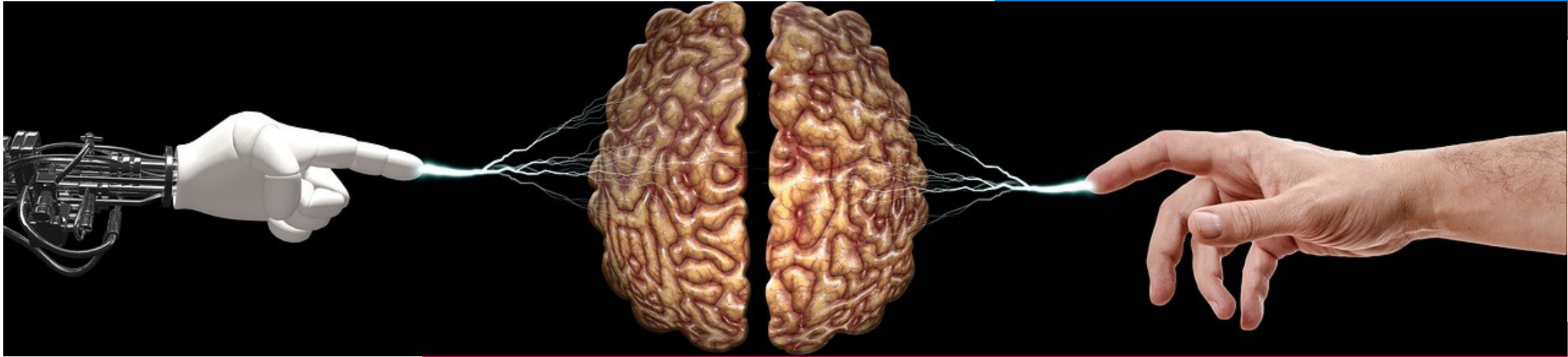
NLP Intro

Almost **6500** languages are spoken in the world.
These are the way to communicate with humans.

Just like these, Humans want to
communicate with Robots or Machines,
& they also want Machines to communicate
with each other.



In short :



NLP is the way to narrow the gap B/W Humans and Robots.

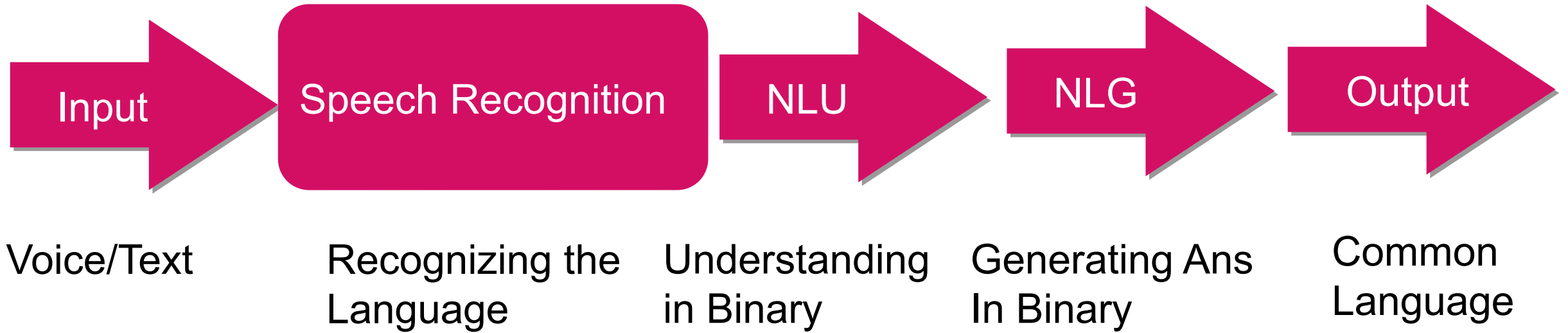
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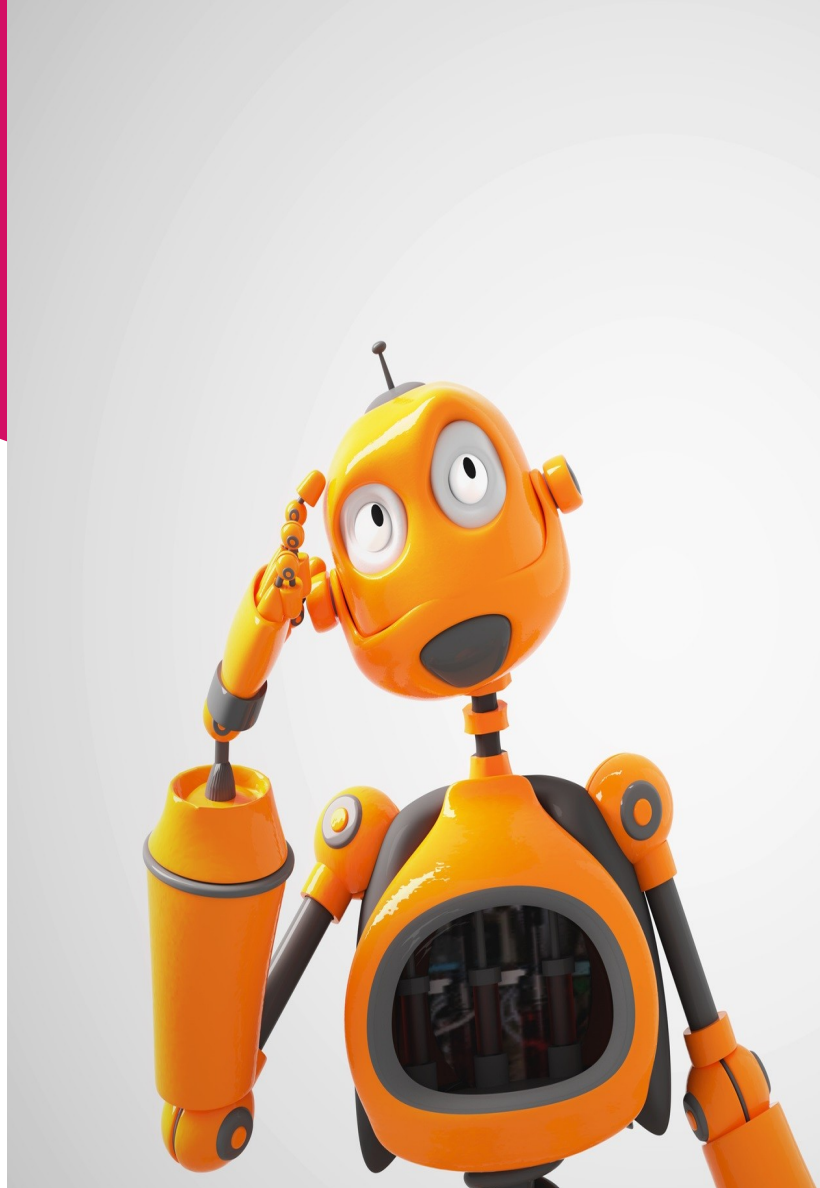


NLP has its roots in 1950s.



How does NLP Work?





NLP Examples

1. Digital Assistants

- ✓ Apple Siri
- ✓ Google Assistant
- ✓ MS Cortona
- ✓ Amazon Alexa
- ✓ Samsung Bixby

GOOGLE

Microsoft

Apple

NLP Examples



2- Speech Reorganization

- ✓ Voice Typing
- ✓ Predictive Text Generation

3- Sentimental Analysis

- ✓ Ai Chatting >> Replika, Waysa
- ✓ Chat Bots >> Eliza, Parry, etc
- ✓ Social Media >> Reactions on posts

4- Advertisements

- ✓ Google, FB, show ads by looking at your search behavior.

NLP Examples



5- Spelling Checking

- ✓ Auto Correction of words while typing >> Grammarly

6- Machine Translation

- ✓ Google Translator

7- Automatic Text Generation

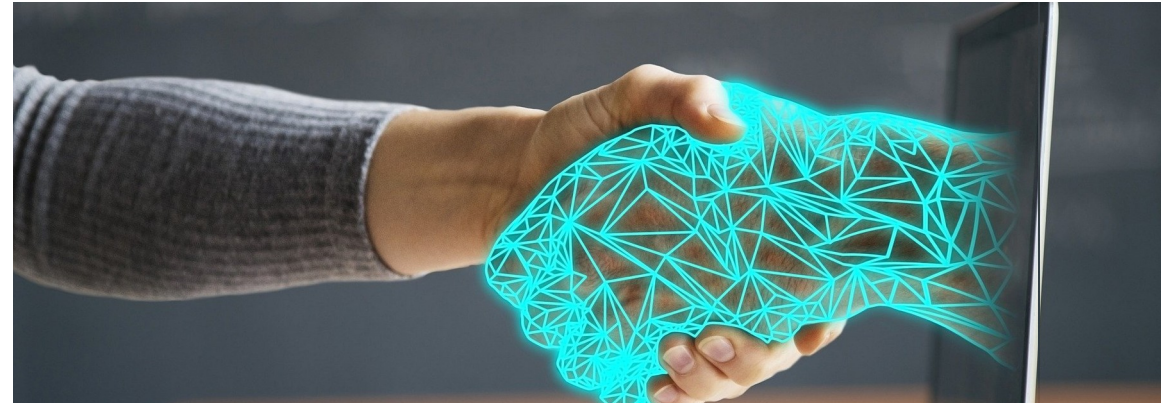
- ✓ Copy.ai (Website)
- ✓ Copyhat (App)

8- QNA

- ✓ Answering to questions directly
- 

Practical Facts

- 1- NLP is a part of Ai*
- 2- NLP focuses on language*
- 3- NLP helps to reduce language barriers*
- 4- NLP is far better than Keyboard Searching*
- 5- NLP is used to know behavior*
- 6- Chat bots are examples of NLP*



Challenges

Various Challenges in NLP

Robot is like a baby. We have to teach it everything like a baby to make it intelligent, fast and accurate.

As NLP is regarding Language, So our main purpose is to teach it Languages.



Challenges in NLP



1- Segmentation

Separation of **sentences** in a paragraph for better understanding.

2- Tokenizing

Separation of **words** in a sentence for better understanding.

3- Stop Words

Removal of **Stop words** (are, and, the, or) for better understanding.

4- Stemming

Giving Machine relevant and same words to understand >> Walk, walking, walked.

5- Lemmatization

Understanding Mood, Behavior, Gender etc

Challenges in NLP

6- POS Tags

Understanding Speech >> Parts of Speech

7- Named Entity Tagging

Telling Machine about famous things,
e.g : famous personalities, Places, Things etc

Google's CEO Sundar Pichai Introduced ***Pixel 6*** in ***New York***.



Drawbacks

- 1- It is very much hard to make a model with the specifications mention above, if there is no any pre-trained model.
- 2- The system may not be able to ans Correc
- 3- There can be Ambiguity. (The ability to understand in more than 1 way.)
- 4- Low Resource languages are not available
- 5- Errors in Text or speech.



Drawbacks / Improvements



Let's Try Google Assistant

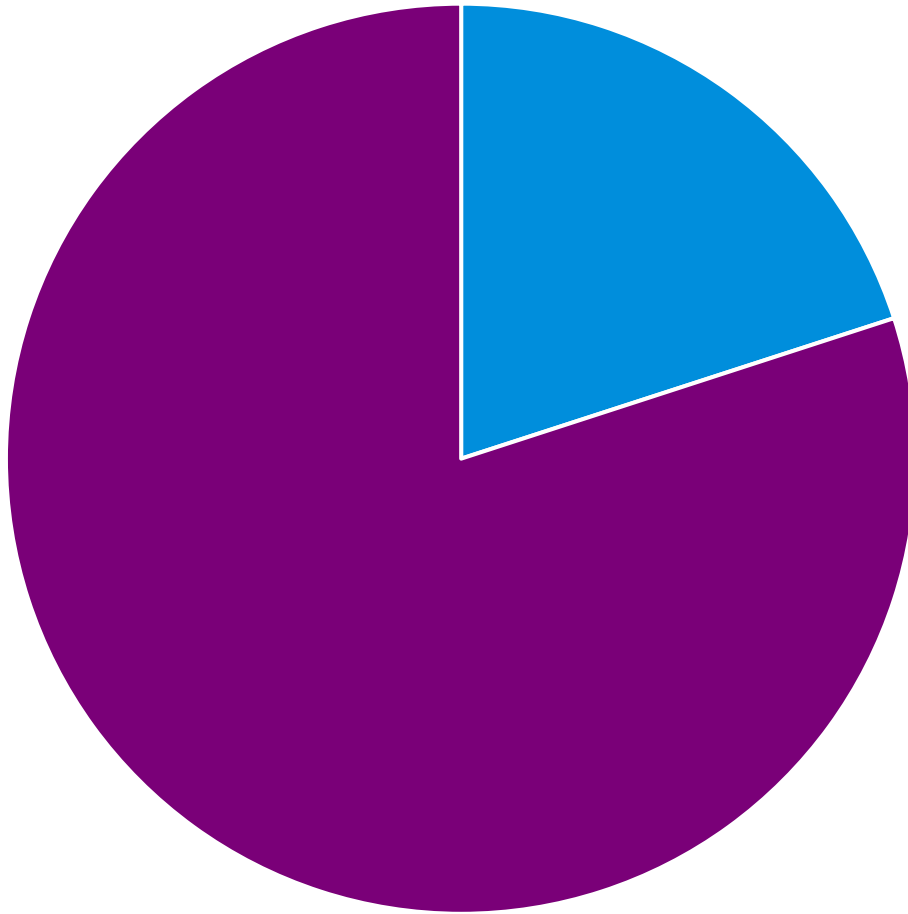


What is my Age?

Do you know my Age?



Data on Internet



■ Structured data ■ Unstructured Data

Improvements

Almost 80% of the data is unstructured on the internet. And only 20-21% is structured.

- 1- If NLP is given more data, it can be improved.
- 2- Information Overloaded

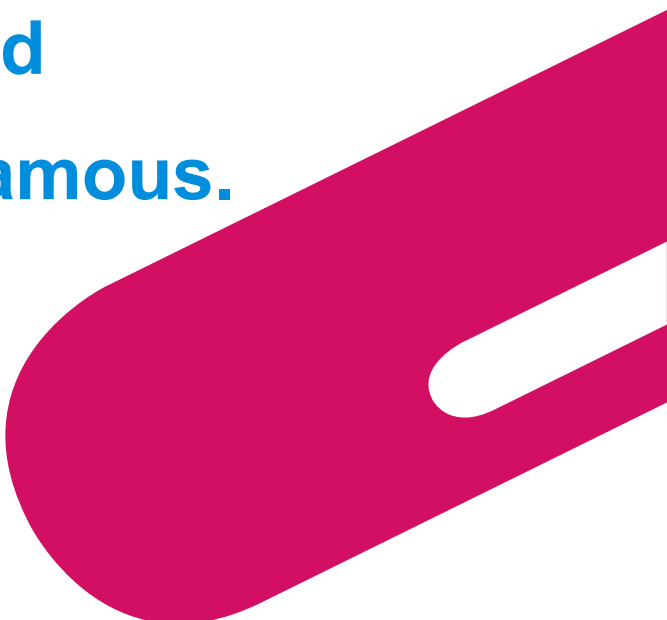
Tools for NLP



NLP is related to Ai, And Ai is a Data Science field
And in Data Science Two Languages are Most Famous.



R & PYTHON



Related Fields



Machine Learning
Artificial Intelligence
Robotics
Neural Networks



Conclusion



Ai, ML, Robotics and many other related fields are nothing if NLP is not present in them.

NLP is the most important factor in Ai. Because it understands languages and then reply with Intelligence inserted in it.

If there is no NLP, Means There is no Ai, Means there are no Ai Applications,

References:

1. YouTube

- Gate Smashers
- Simplilearn
- IBM Technology
- Edurekha

2. Websites

- Wikipedia

